



Wexford County Landfill

Manton, Michigan

Surface and Penetration Emission Monitoring 2nd Quarter Report 2025

Prepared By:



39365 W. 12 Mile Road
Farmington Hills, MI 48331
(877) 633-5520



June 10, 2025

Mr. Anthony Testa
Regional Landfill Gas Program Manager
Wexford County Landfill
990 N Mackinaw Trail
Manton, MI 48457

RE: **Surface and Penetration Emission Monitoring (SEM) Report – 2nd Quarter 2025
Wexford County Landfill, Manton, Michigan**

Dear Mr. Testa,

Tetra Tech has prepared the enclosed report documenting the results of the 2nd Quarter 2025 surface and penetration emission monitoring (SEM) at Wexford County Landfill in Manton, Michigan.

The initial monitoring event was performed on May 5, 2025. Results of the scan indicated one (1) location exhibiting methane (CH₄) concentrations greater than the 500 parts per million by volume (ppm) above background regulatory standard.

The 10-day recheck event was performed on May 14, 2025. The one (1) exceedance location exhibited CH₄ concentrations of less than 500ppm above background and were scheduled for one-month (30-day) remonitoring.

The one month (30-day) recheck event was performed on June 3, 2025. The exceedance location exhibited CH₄ concentrations of less than 500ppm above background.

In summary, one (1) of one (1) exceedance locations were resolved by corrective action during the 2nd Quarter 2025 SEM. No further monitoring is required for this event.

Below is the surface scan exceedance summary table:

Exceedance ID	Latitude	Longitude	Initial Scan	10-Day	30-Day	Final Result
#1 Surface	44.351231	-85.3946	5/5/2025 913ppm	5/14/2025 ppm	6/3/2025 72ppm	PASS

Weather conditions recorded during the monitoring events were as follows:

May 5, 2025

- Temperature approximately 55° F
- Relative humidity of 70 percent
- Barometric pressure of 30.04" Hg
- Wind E at 12 mph
- Overcast

May 14, 2025

- Temperature approximately 63° F
- Relative humidity of 86 percent
- Barometric pressure of 29.8" H
- Wind SE at 5 mph
- Overcast, drizzle

June 3, 2025

- Temperature approximately 70° F
- Relative humidity of 63 percent
- Barometric pressure of 29.85" Hg
- Wind S at 13 mph
- Sunny

The SEM procedures were conducted as required by the United States Environmental Protection Agency (USEPA) in Federal Plan Subpart OOO (Subpart OOO), 40 CFR 63.1960 (c) and (d), the National Emission Standard for Hazardous Air Pollutants (NESHAP), 40 CFR 63.1958 (d) and 63.1960 (c), and 40 CFR 60, Appendix A, Method 21. This monitoring event was performed during typical meteorological conditions.

Facility personnel used an Elkins Earthworks Inficon IRwin Portable Methane Detector (IRwin) to perform the emissions monitoring. During the event, attention was given to monitoring unusual cover conditions (stressed vegetation, cracks, seeps, etc.) and areas with unusual odors. The IRwin was calibrated at the beginning of each day, prior to performing the monitoring, in accordance with Method 21 compliance requirements. Upwind and downwind readings were taken beyond the landfill limits to establish background concentrations prior to each day's monitoring. Calibration logs were completed by the field technician performing the work, and are included in Appendix A. During the monitoring event, Tetra Tech observed that the ground surface appeared to be in good condition overall and there were no unusual odors noted. Results are presented in the attached forms.

If you have any questions regarding this report, please contact me at (989) 351-9134.

Sincerely,



Jeremy West
Project Manager
Tetra Tech

Encl: Appendix A – Daily Calibration Logs
Appendix B - Daily Surface Monitoring Logs
Appendix C - Exceedance & Monitoring Route Maps
Appendix D - Calibration Gas Certifications

cc: Tami Craig – GFL
David Knapp, Maria Work – Tetra Tech (electronically)

Tetra Tech – Cornerstone Environmental Group
39395 W 12 Mile Rd Ste 200
Farmington Hills, MI 48331
<http://www.cornerstoneeg.com>

Wexford County Landfill

Manton, Michigan

Appendix A

Daily Calibration Logs



PART 1 CALIBRATION PRECISION TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 Initial SEM

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:57 AM

DATE: May 5, 2025

Calibration Gas Standard: 500 ppm (7) (check cal. gas cert. - should be 500 ppm)

MEASUREMENT # 1:

Meter Reading for Zero Air: 0.0 ppm (1)

Meter Reading for Calibration Gas: 489.0 ppm (2)

MEASUREMENT # 2:

Meter Reading for Zero Air: 0.0 ppm (3)

Meter Reading for Calibration Gas: 488.0 ppm (4)

MEASUREMENT # 3:

Meter Reading for Zero Air: 0.0 ppm (5)

Meter Reading for Calibration Gas: 489.0 ppm (6)

CALCULATE PRECISION:

Must be less than 10%

$$\frac{|(7) - (2)| + |(7) - (4)| + |(7) - (6)|}{3} \times \frac{1}{(7)} \times \frac{100}{1} = \underline{2.267\%}$$

CALIBRATION GAS CERTIFICATION DATA AND EXPIRATION DATE:

Zero Gas Serial Number: 012224YG-C

Span Gas Serial Number: 020524YG-X

Zero Gas Expiration Date: 01/24/26

Span Gas Expiration Date: 02/05/26

INCLUDE A COPY OF THE CALIBRATION GAS CERTIFICATION SHEET FROM GAS SUPPLIER/MANUFACTURER



PART 2 INSTRUMENT RESPONSE TIME TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 Initial SEM

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:57 AM

DATE: May 5, 2025

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 489.0 ppm

90% of the Stabilized Reading: 440.1 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 5.0 seconds (1)

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 488.0 ppm

90% of the Stabilized Reading: 439.2 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (2)

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 489.0 ppm

90% of the Stabilized Reading: 440.1 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (3)

CALCULATE RESPONSE TIME:

Must be less than 30 seconds

$$\frac{(1) + (2) + (3)}{3}$$

=

3.667 seconds



PART 3

STABILIZED READING AND BACKGROUND DETERMINATION

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 Initial SEM

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:57 AM

DATE: May 5, 2025

STABILIZED READING DETERMINATION PROCEDURE

Calibration Gas Standard: 500.0

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 489.0 ppm

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 488.0 ppm

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 489.0 ppm

Stable Instrument Reading:
$$\frac{\text{Measurement \#1} + \text{Measurement \#2} + \text{Measurement \#3}}{3}$$
$$= \underline{488.7} \text{ ppm}$$

BACKGROUND DETERMINATION PROCEDURE:

1. Upwind Reading (highest in 30 seconds): 0 ppm (1)

2. Downwind Reading (highest in 30 seconds): 4 ppm (2)

Calculate Background Value:
$$\frac{(1) + (2)}{2} = \underline{2.0} \text{ ppm}$$



PART 1 CALIBRATION PRECISION TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 10-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:12 AM

DATE: May 14, 2025

Calibration Gas Standard: 500 ppm (7) (check cal. gas cert. - should be 500 ppm)

MEASUREMENT # 1:

Meter Reading for Zero Air: 0.0 ppm (1)

Meter Reading for Calibration Gas: 466.0 ppm (2)

MEASUREMENT # 2:

Meter Reading for Zero Air: 0.0 ppm (3)

Meter Reading for Calibration Gas: 463.0 ppm (4)

MEASUREMENT # 3:

Meter Reading for Zero Air: 0.0 ppm (5)

Meter Reading for Calibration Gas: 462.0 ppm (6)

CALCULATE PRECISION:

Must be less than 10%

$$\frac{|(7) - (2)| + |(7) - (4)| + |(7) - (6)|}{3} \times \frac{1}{(7)} \times \frac{100}{1} = \underline{7.267\%}$$

CALIBRATION GAS CERTIFICATION DATA AND EXPIRATION DATE:

Zero Gas Serial Number: 012224YG-C

Span Gas Serial Number: 020524YG-X

Zero Gas Expiration Date: 01/24/26

Span Gas Expiration Date: 02/05/26

INCLUDE A COPY OF THE CALIBRATION GAS CERTIFICATION SHEET FROM GAS SUPPLIER/MANUFACTURER



PART 2

INSTRUMENT RESPONSE TIME TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 10-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:12 AM

DATE: May 14, 2025

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 466.0 ppm

90% of the Stabilized Reading: 419.4 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 5.0 seconds (1)

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 463.0 ppm

90% of the Stabilized Reading: 416.7 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (2)

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 462.0 ppm

90% of the Stabilized Reading: 415.8 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (3)

CALCULATE RESPONSE TIME:

Must be less than 30 seconds

$$\frac{(1) + (2) + (3)}{3}$$

=

3.667 seconds



PART 3

STABILIZED READING AND BACKGROUND DETERMINATION

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 10-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:12 AM

DATE: May 14, 2025

STABILIZED READING DETERMINATION PROCEDURE

Calibration Gas Standard: 500.0

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 466.0 ppm

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 463.0 ppm

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 462.0 ppm

Stable Instrument Reading:
$$\frac{\text{Measurement \#1} + \text{Measurement \#2} + \text{Measurement \#3}}{3}$$
$$= \underline{463.7} \text{ ppm}$$

BACKGROUND DETERMINATION PROCEDURE:

1. Upwind Reading (highest in 30 seconds): 0 ppm (1)

2. Downwind Reading (highest in 30 seconds): 0 ppm (2)

Calculate Background Value:
$$\frac{(1) + (2)}{2} = \underline{0.0} \text{ ppm}$$



PART 1 CALIBRATION PRECISION TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 30-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:44 AM

DATE: June 3, 2025

Calibration Gas Standard: 500 ppm (7) (check cal. gas cert. - should be 500 ppm)

MEASUREMENT # 1:

Meter Reading for Zero Air: 0.0 ppm (1)

Meter Reading for Calibration Gas: 492.0 ppm (2)

MEASUREMENT # 2:

Meter Reading for Zero Air: 0.0 ppm (3)

Meter Reading for Calibration Gas: 487.0 ppm (4)

MEASUREMENT # 3:

Meter Reading for Zero Air: 0.0 ppm (5)

Meter Reading for Calibration Gas: 487.0 ppm (6)

CALCULATE PRECISION:

Must be less than 10%

$$\frac{|(7) - (2)| + |(7) - (4)| + |(7) - (6)|}{3} \times \frac{1}{(7)} \times \frac{100}{1} = \underline{2.267\%}$$

CALIBRATION GAS CERTIFICATION DATA AND EXPIRATION DATE:

Zero Gas Serial Number: 012224YG-C

Span Gas Serial Number: 020524YG-X

Zero Gas Expiration Date: 01/24/26

Span Gas Expiration Date: 02/05/26

INCLUDE A COPY OF THE CALIBRATION GAS CERTIFICATION SHEET FROM GAS SUPPLIER/MANUFACTURER



PART 2 INSTRUMENT RESPONSE TIME TEST RECORD

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 30-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:44 AM

DATE: June 3, 2025

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 492.0 ppm

90% of the Stabilized Reading: 442.8 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 5.0 seconds (1)

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 487.0 ppm

90% of the Stabilized Reading: 438.3 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (2)

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 487.0 ppm

90% of the Stabilized Reading: 438.3 ppm

Time to Reach 90% of Stabilized reading after switching from
Zero Air to Calibration Gas: 3.0 seconds (3)

CALCULATE RESPONSE TIME:

Must be less than 30 seconds

$$\frac{(1) + (2) + (3)}{3}$$

=

3.667 seconds



PART 3

STABILIZED READING AND BACKGROUND DETERMINATION

Initial Event: May 5, 2025

LANDFILL NAME: Wexford Landfill

EVENT: Q2 2025 30-Day SEM Recheck

INSTRUMENT MAKE: Inficon

MODEL: IRwin

SERIAL #: 92006184

PERFORMED BY: Titus Camero

TIME: 9:44 AM

DATE: June 3, 2025

STABILIZED READING DETERMINATION PROCEDURE

Calibration Gas Standard: 500.0

MEASUREMENT # 1:

Stabilized Reading Using Calibration Gas: 492.0 ppm

MEASUREMENT # 2:

Stabilized Reading Using Calibration Gas: 487.0 ppm

MEASUREMENT # 3:

Stabilized Reading Using Calibration Gas: 487.0 ppm

Stable Instrument Reading:
$$\frac{\text{Measurement \#1} + \text{Measurement \#2} + \text{Measurement \#3}}{3}$$
$$= \underline{488.7} \text{ ppm}$$

BACKGROUND DETERMINATION PROCEDURE:

1. Upwind Reading (highest in 30 seconds): 1 ppm (1)

2. Downwind Reading (highest in 30 seconds): 11 ppm (2)

Calculate Background Value:
$$\frac{(1) + (2)}{2} = \underline{6.0} \text{ ppm}$$

Wexford County Landfill

Manton, Michigan

Appendix B

Daily Exceedance Monitoring Logs

Individual Monitoring Exceedance Surface Monitoring Design Plan

Exceedance #: 1

Date: <u>5-5-2025</u>	Time: <u>11:41 AM</u>	Monitoring Technician Initials: <u>TC</u>
Instrument reading - Background reading: <u>913.0</u> ppm - <u>2.0</u> ppm = <u>911.0</u> ppm		
Location of monitored exceedance (include description of field marker used): <u>15 yd W of stake 532, N slope</u>		
Location marked with orange pin flag. <u>44.35123</u> <u>-85.39460</u>		
Describe cover maintenance or adjustments to the vacuum of adjacent wells to increase gas collection in vicinity of measured exceedance before remonitoring in 10 days: <u>Added clay/dirt or hydrated bentonite around area.</u>		

Remonitor location within 10 calendar days of Initial Exceedance:

Date: <u>5-14-2025</u>	Time: <u>9:20 AM</u>	Monitoring Technician Initials: <u>TC</u>
Instrument reading - Background reading: <u>341.0</u> ppm - <u>0.0</u> ppm = <u>341.0</u> ppm		
If the 10-day remonitoring shows an exceedance, describe additional corrective action taken before remonitoring again within 10 days: _____		
If the 10-day remonitoring is <500 ppm, remonitor 1 month from Initial Exceedance:		
Date: <u>6-3-2025</u>	Time: <u>9:55 AM</u>	Monitoring Technician Initials: <u>TC</u>
Instrument reading - Background reading: <u>72.0</u> ppm - <u>6.0</u> ppm = <u>66.0</u> ppm		
If the 1-month remonitoring is <500 ppm, resume normal quarterly monitoring.		
If the 1-month remonitoring shows an exceedance, describe additional corrective action taken before remonitoring again within 10 Days: _____		

Remonitor location within 10 calendar days of 2nd Exceedance:

Date: _____	Time: _____	Monitoring Technician Initials: _____
Instrument reading - Background reading: _____ ppm - _____ ppm = _____ ppm		
If the 10-day remonitoring is <500 ppm, remonitor 1 month from Initial Exceedance:		
Date: _____	Time: _____	Monitoring Technician Initials: _____
Instrument reading - Background reading: _____ ppm - _____ ppm = _____ ppm		

If the 1-month remonitoring is <500 ppm, resume normal monitoring.

If the 1-month remonitoring shows an exceedance, describe additional corrective action taken before remonitoring again within 10 Days: _____

Use additional forms if necessary

If monitoring shows 3 exceedances within a quarterly period, a new well or other collection device must be installed within 120 Days of the initial exceedance or alternative remedies/timelines may be submitted to the Administrator for approval. Further monitoring is not necessary until the remedy is completed. The 3 exceedances do NOT have to be consecutive.

Use this form to record an individual monitoring exceedance and follow-up monitoring activities.

This form is only to be used when a reading of 500 ppm above background is encountered during the surface monitoring.

Use a separate form for each initial exceedance.

Wexford County Landfill

Manton, Michigan

Appendix C

Topographic Exceedance and Monitoring Route Maps

Wexford Landfill

Q2 2025 SEM

Legend

▲ Surface Exceedance

▲ Ex 1- 913ppm

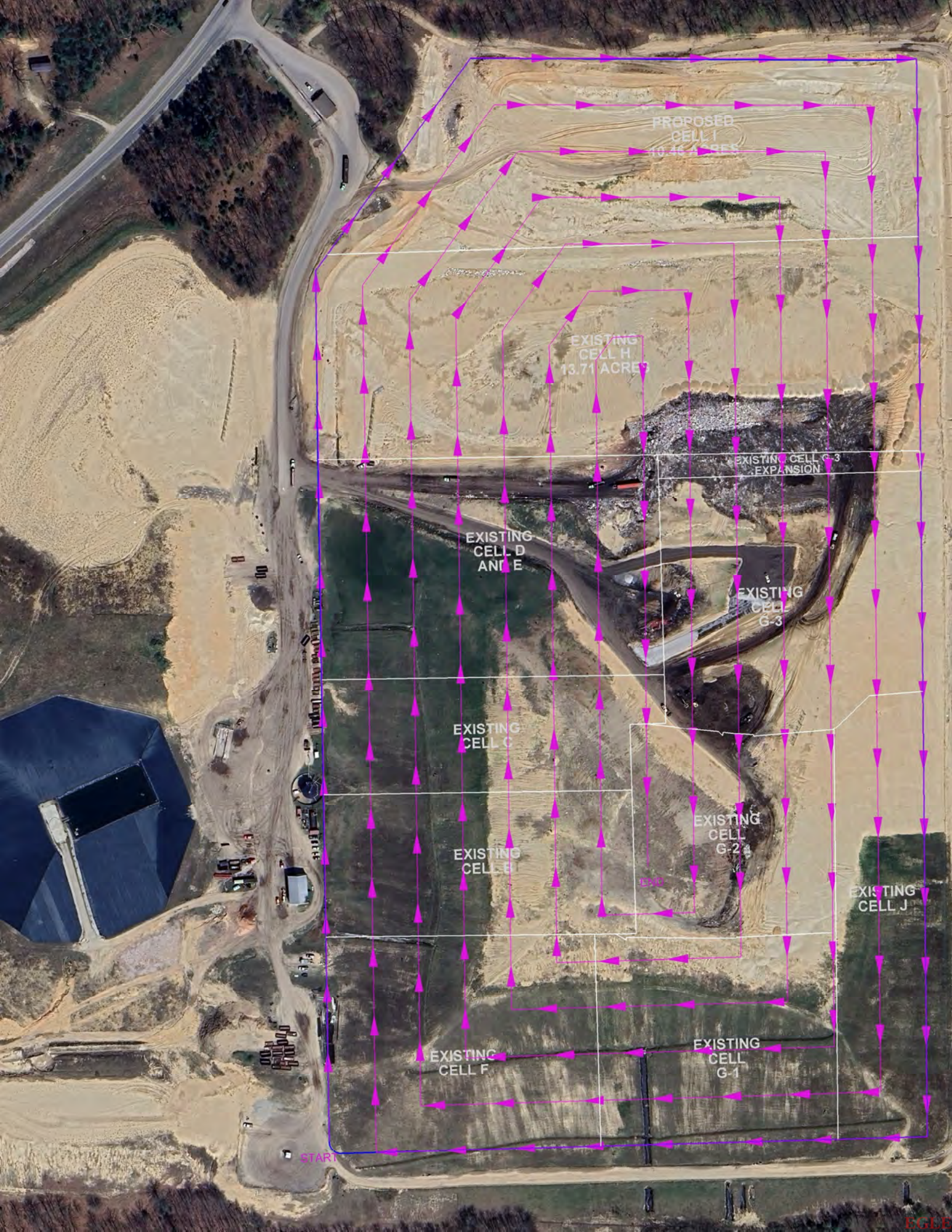
Google Earth

Image © 2025 Airbus



1000 ft

EGLE



PROPOSED
CELL I
10.46 ACRES

EXISTING
CELL H
13.71 ACRES

EXISTING CELL G-3
EXPANSION

EXISTING
CELL
G-3

EXISTING
CELL D
AND E

EXISTING
CELL C

EXISTING
CELL E

EXISTING
CELL
G-2

EXISTING
CELL F

EXISTING
CELL
G-1

EXISTING
CELL J

START

EGLE

Wexford County Landfill

Manton, Michigan

Appendix D

Calibration Gas Certifications



GASCO AFFILIATES, LLC.

320 Scarlet Blvd.
Oldsmar, FL 34677
(800) 910-0051
fax: (866) 755-8920
www.gascogas.com

CERTIFICATE OF ANALYSIS

Date: May 3, 2022
Order Number: 3344
Lot Number: 304-402287502-1

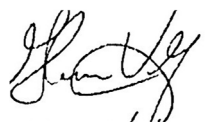
Customer: Pangaea Gases LLC

Use Before: 05/03/2026

Component	Requested Concentration	Analytical Result (+/- 2%)
Methane Air	500 PPM Balance	549 PPM Balance
Cylinder Size: 3.70 Cu. Ft. Contents: 105 Liter		Valve: 5/8"-18 UNF Pressure: 1200 psig

Product composition verified by direct comparison to calibration standards traceable to N.I.S.T. weights and/ or N.I.S.T. Gas Mixture reference materials.

Analyst:


Glenn Velez



GASCO AFFILIATES, LLC.

320 Scarlet Blvd.
Oldsmar, FL 34677
(800) 910-0051
fax: (866) 755-8920
www.gascogas.com

CERTIFICATE OF ANALYSIS

Date: May 3, 2022
Order Number: 3344
Lot Number: 304-402433798-1

Customer: Pangaea Gases LLC

Use Before: 05/03/2026

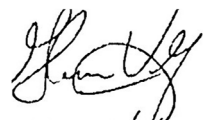
Component	Requested Concentration	Analytical Result
Oxygen	20-22% vol.	20.9% vol. THC <0.1 PPM

Cylinder Size: 3.70 Cu. Ft.
Contents: 105 Liter

Valve: 5/8"-18 UNF
Pressure: 1200 psig

Impurities verified against analytical standards traceable to NIST by weight and/or analysis.

Analyst:


Glenn Velez



CERTIFICATE OF ANALYSIS
CERTIFIED STANDARD
Pangaea Gases, LLC. PO Box 404 Burton, Ohio 44021

Customer: Tetra Tech
CGA: C-10
Customer PO #: 3520
Cylinder Style: 103L NRC
Part #: SDI-2077C

Lot #: 100523YG-E
Certification Date: 10/05/2023
Expiration Date: 10/05/2025
Pressure, psig: 1000

Components	Requested Concentration	Certified Concentration	Analytical Accuracy
Methane	500 ppm	505 ppm	± 5%
Air	Balance	Balance	-

This report states accurately the results of the investigation made upon the material submitted to the analytical laboratory. Every effort has been made to determine objectively the information requested. However, in connection with this report, Global Calibration Gases LLC shall have no liability in excess of the established charge for this service. Produced at Global Calibration Gases LLC, Sarasota, Florida.